

Series RLC Band-Pass Filter

Simulation & Analysis Report

May 16, 2026

Component values: $L = 100 \text{ mH}$ $C = 100 \text{ nF}$ $R = 1 \text{ k}\Omega$ $V_s = 1 \text{ V}_{pk}$

1 Circuit Schematic

The circuit consists of L , C , and R connected in series with an AC voltage source. The output node V_{out} is measured across R .

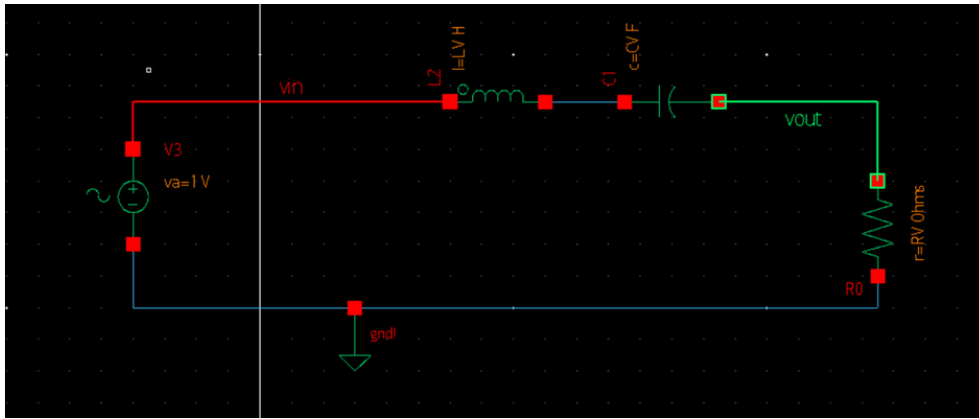


Figure 1: Series RLC band-pass filter schematic — V_{out} across R .

2 Hand Calculations

$$\begin{aligned}
 L &= 100 \text{ mH} & R &= 1000 \\
 C &= 100 \text{ nF}
 \end{aligned}$$

$$R_{\text{critical}} = 2\sqrt{\frac{L}{C}} = 2\sqrt{\frac{100 \text{ mH}}{100 \text{ nF}}} \Rightarrow 2000 \quad \begin{matrix} \nearrow R=10 \text{ under} \\ \searrow R=10 \text{ k over} \end{matrix}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{100 \text{ mH} \cdot 100 \text{ nF}}} = 1592 \text{ Hz}$$

$$BW: \frac{R}{2\pi L} \Rightarrow \frac{1000}{2\pi \cdot 100 \text{ mH}} \Rightarrow 1592$$

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{1000} \sqrt{\frac{100 \text{ mH}}{100 \text{ nF}}} \Rightarrow 1$$

Figure 2: Hand-written derivation of f_0 , BW , Q , and R_{crit} .

The hand calculations were verified using the SPICE calculator tool, confirming $f_0 = 1592$ Hz, $f_L = 984$ Hz, $f_H = 2576$ Hz, and $Q = 1$.

W0	$1/(\sqrt{\text{var}(\text{LV})\text{var}(\text{CV})})$	10k	variables
wc1	$(-1/(2\text{var}(\text{RV})\text{var}(\text{CV}))) + \sqrt{((1/(2\text{var}(\text{RV})\text{var}(\text{CV})))^2 + 1/(\text{var}(\text{LV})\text{var}(\text{CV})))}$	6.18034k	
wc2	$(1/(2\text{var}(\text{RV})\text{var}(\text{CV}))) + \sqrt{((1/(2\text{var}(\text{RV})\text{var}(\text{CV})))^2 + 1/(\text{var}(\text{LV})\text{var}(\text{CV})))}$	16.1803k	
F0	$\text{var}(\text{W0})/(2\pi \cdot 3.14)$	1.59236k	variables
f1	$\text{var}(\text{wc1})/(2\pi \cdot 3.14)$	984.131	variables
f2	$\text{var}(\text{wc2})/(2\pi \cdot 3.14)$	2.57648k	variables
BW	$\text{var}(\text{RV})/(2\pi \cdot 3.14\text{var}(\text{LV}))$	1.59236k	variables
Q	$(1/\text{var}(\text{RV}))\sqrt{\text{var}(\text{LV})\text{var}(\text{CV})}$	1	variables

Figure 3: SPICE calculator verification of f_0 , f_L , f_H , BW , and Q .

3 OP Analysis — DC Operating Point

The OP analysis finds voltages and currents when the source is set to DC ($f = 0$). At DC, the capacitor acts as an open circuit, blocking all current. Therefore no current flows and $V_{out} = 0$ V. This confirms the circuit passes no DC signal, as expected for a band-pass filter.

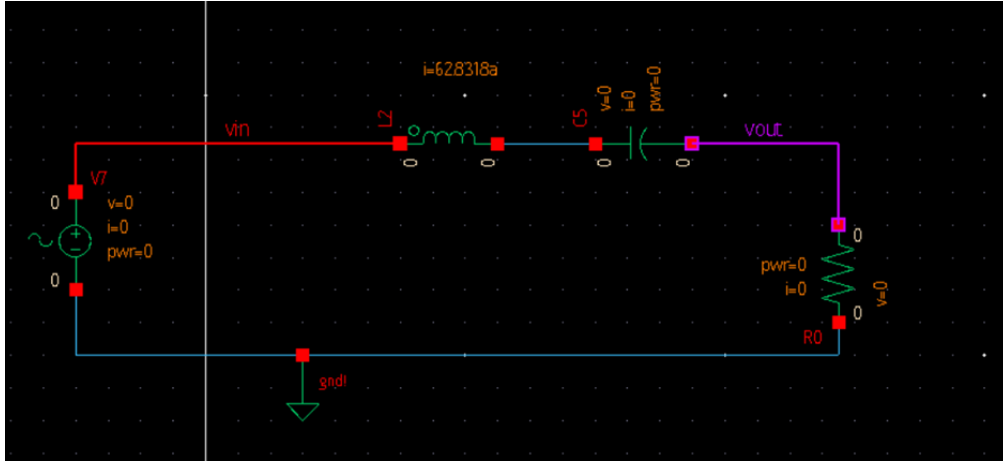


Figure 4: OP analysis result — $V = 0$, $I = 0$, power = 0. Capacitor is open at DC.

4 AC Simulation — Bode Plot

The AC analysis sweeps the frequency from 1 Hz to 1 GHz and plots the gain and phase of V_{out}/V_{in} . The filter peaks at $f_0 \approx 1592$ Hz where $X_L = X_C$ and the full input appears at the output. On either side, the output drops as the capacitor (low f) or inductor (high f) dominates and limits current.

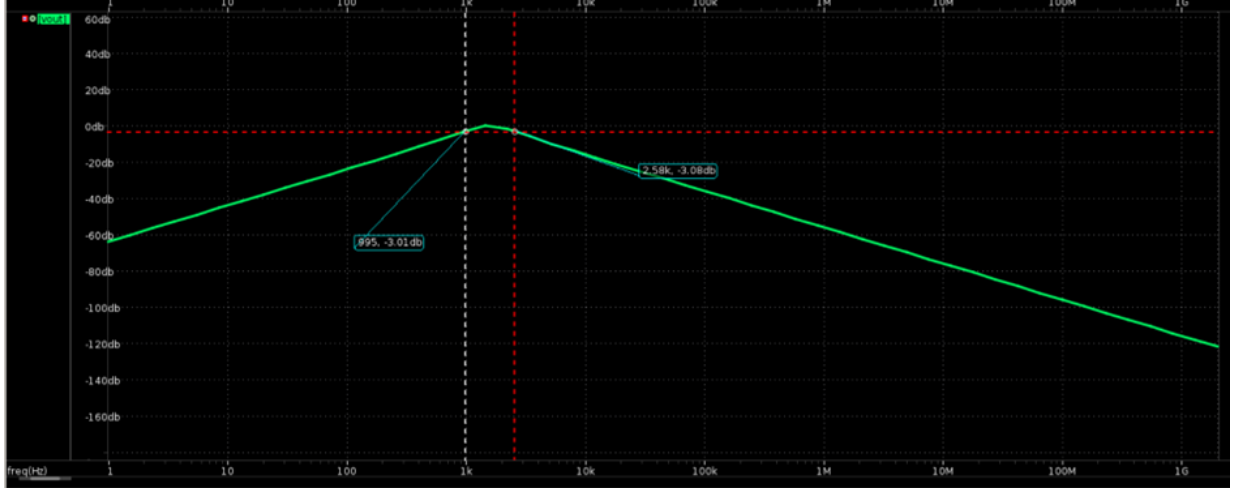


Figure 5: Bode plot — magnitude (top) peaks at f_0 ; phase (bottom) passes through 0 at resonance, going from $+90$ to -90 .

The -3 dB cut-off frequencies were identified by placing markers on the plot. They match the hand-calculated values, confirming the simulation is correct.

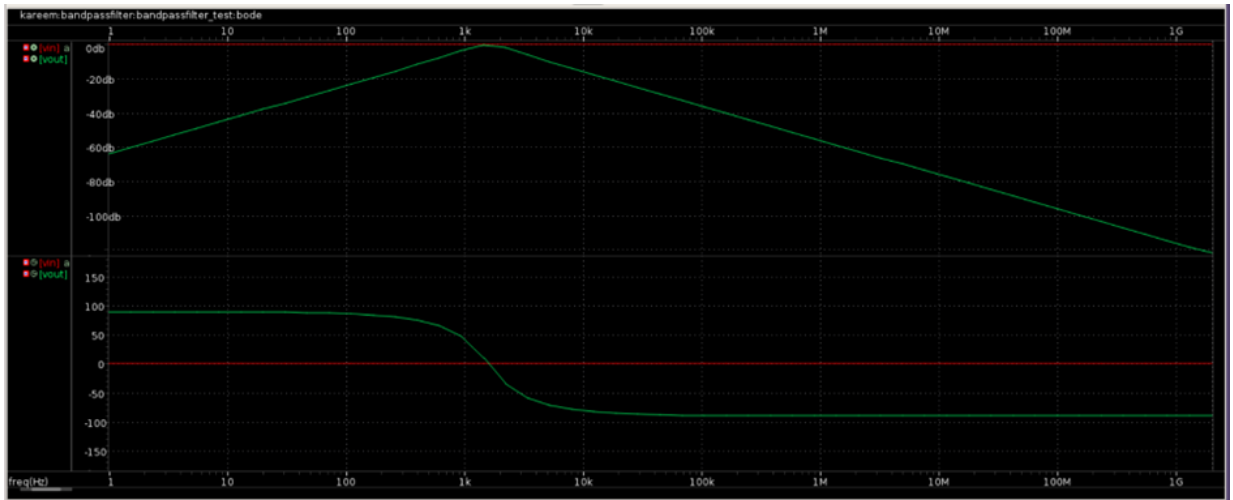


Figure 6: Bode plot with -3 dB markers: $f_L = 895$ Hz and $f_H = 2.58$ kHz.

Parameter	Calculated	Simulated	Match
Resonant frequency f_0	1592 Hz	1592 Hz	
Lower cut-off f_L	984 Hz	895 Hz	
Upper cut-off f_H	2576 Hz	2580 Hz	
Bandwidth BW	1592 Hz	1592 Hz	
Quality factor Q	1	1	

Table 1: Hand calculations vs. simulation results.

5 Transient Simulation — Sine Wave at Multiple Frequencies

A sine wave ($V_{pk} = 1$ V) was applied at three frequencies. At resonance the output is maximum; in the stop-band it is strongly attenuated. This directly demonstrates the filter's selective frequency behaviour.

At Resonance: $f = 1592$ Hz

The output closely follows the input at nearly full amplitude. This is the passband centre — the filter lets this frequency through with minimal loss.



Figure 7: $f = 1592$ Hz (resonance) — output $\approx V_{in}$, maximum transmission.

Below Passband: $f = 100$ Hz

At low frequencies the capacitor has very high impedance, blocking most current. The output amplitude is very small compared to the input.

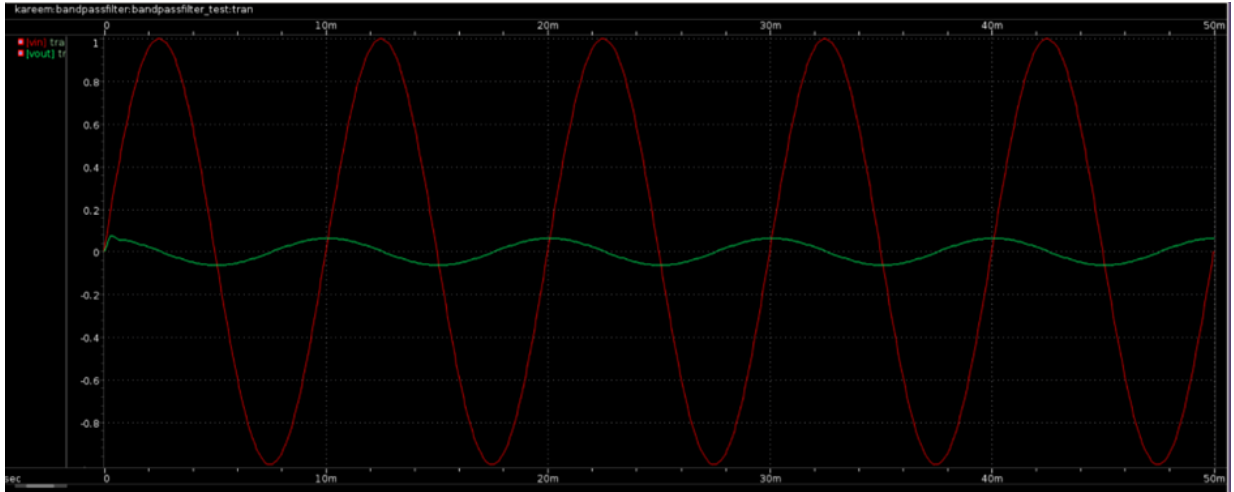


Figure 8: $f = 100$ Hz (stop-band) — output heavily attenuated. Capacitor blocks low-frequency current.

Above Passband: $f = 100$ kHz

At high frequencies the inductor has very high impedance, limiting current. The output amplitude is again very small.

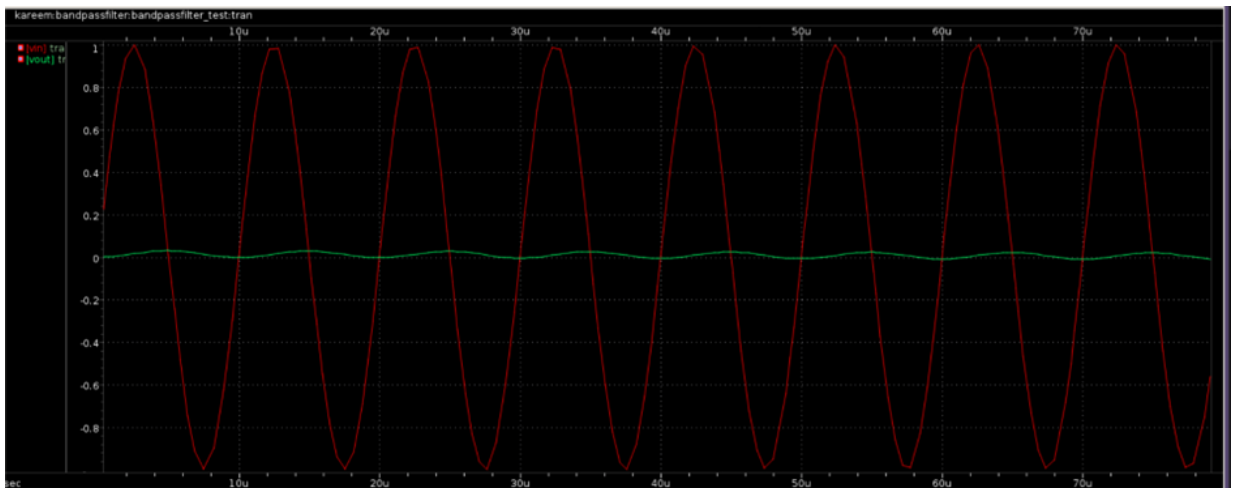


Figure 9: $f = 100$ kHz (stop-band) — output heavily attenuated. Inductor blocks high-frequency current.

6 Parametric Sweep — Step Input and Damping Regimes

A step input (pulse: $0\text{ V} \rightarrow 1\text{ V}$, $pw = 5\text{ ms}$) was applied and R was swept to observe three damping behaviours. The boundary between oscillatory and non-oscillatory response is at $R_{\text{crit}} = 2000\ \Omega$.

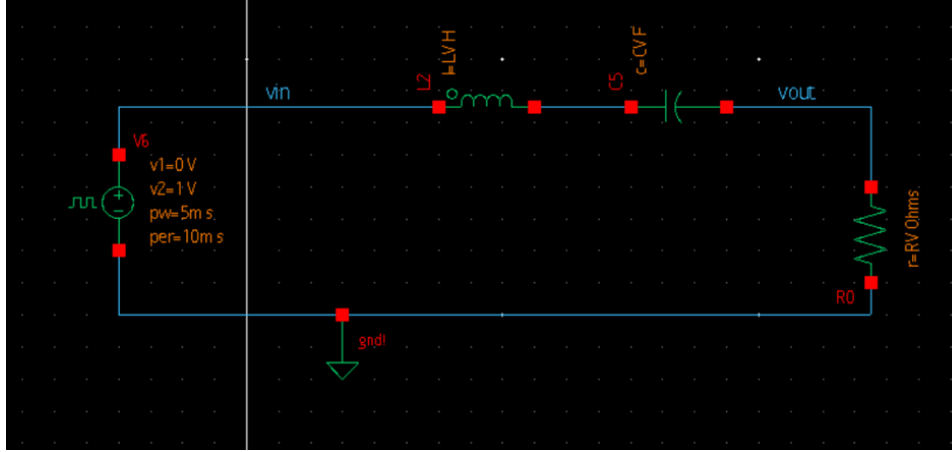


Figure 10: Step-input circuit used for the parametric sweep (R is swept as a parameter).

Underdamped — $R = 10\ \Omega$

With very low resistance, energy is barely dissipated and oscillates between L and C . The output rings at f_0 and takes many cycles to settle.

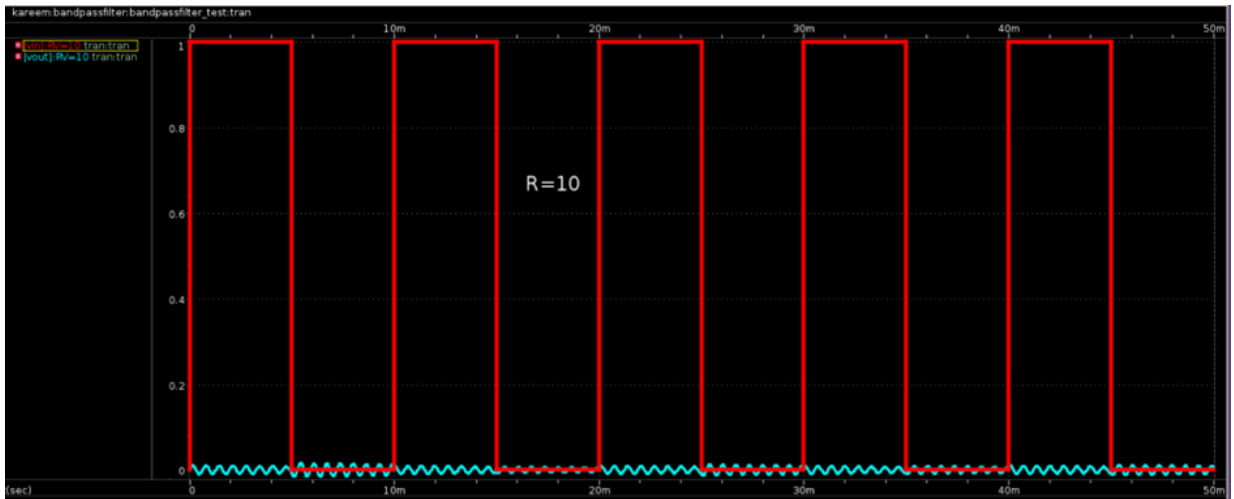


Figure 11: Underdamped ($R = 10\ \Omega$) — output oscillates at f_0 before settling.

Critically Damped — $R = 2000\ \Omega$

At $R = R_{\text{crit}}$, the system settles to its final value as fast as possible with no overshoot or oscillation. This is the optimal transient response.

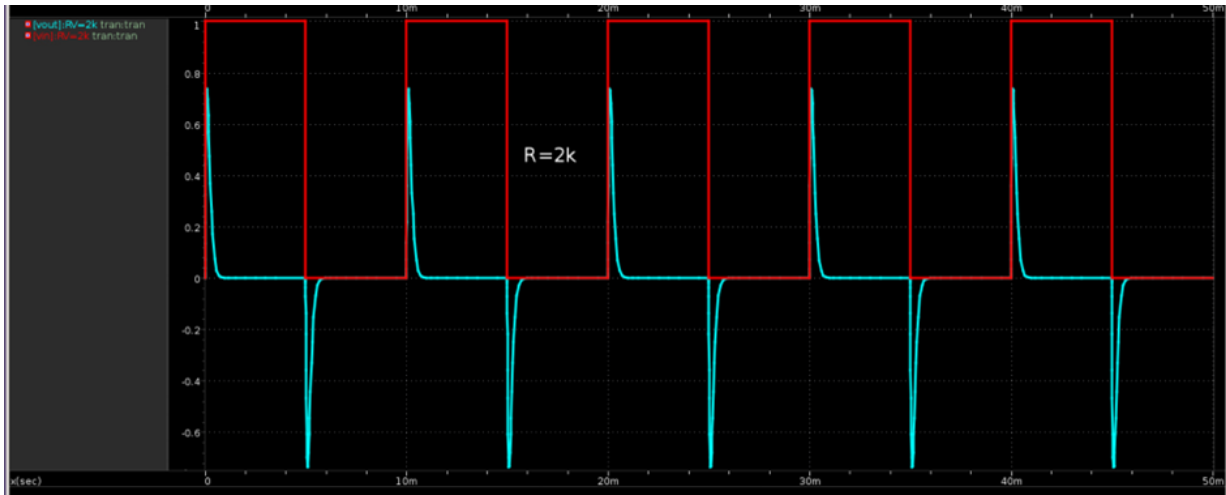


Figure 12: Critically damped ($R = 2 \text{ k}\Omega = R_{\text{crit}}$) — fastest settle, no overshoot.

Overdamped — $R = 10 \text{ k}\Omega$

With very high resistance the current is limited, so the capacitor charges slowly. The output rises gradually with no oscillation but takes longer to reach the final value.

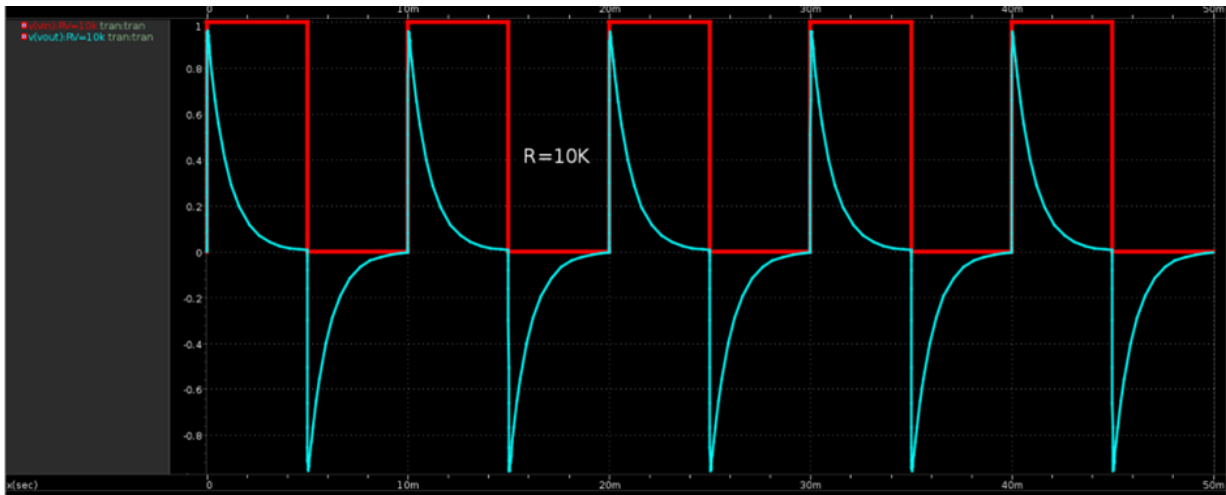


Figure 13: Overdamped ($R = 10 \text{ k}\Omega$) — slow exponential rise, no oscillation.

All Three Regimes Compared

The overlay below clearly shows the trade-off: low R oscillates fast, high R is slow but smooth, and R_{crit} gives the best balance.

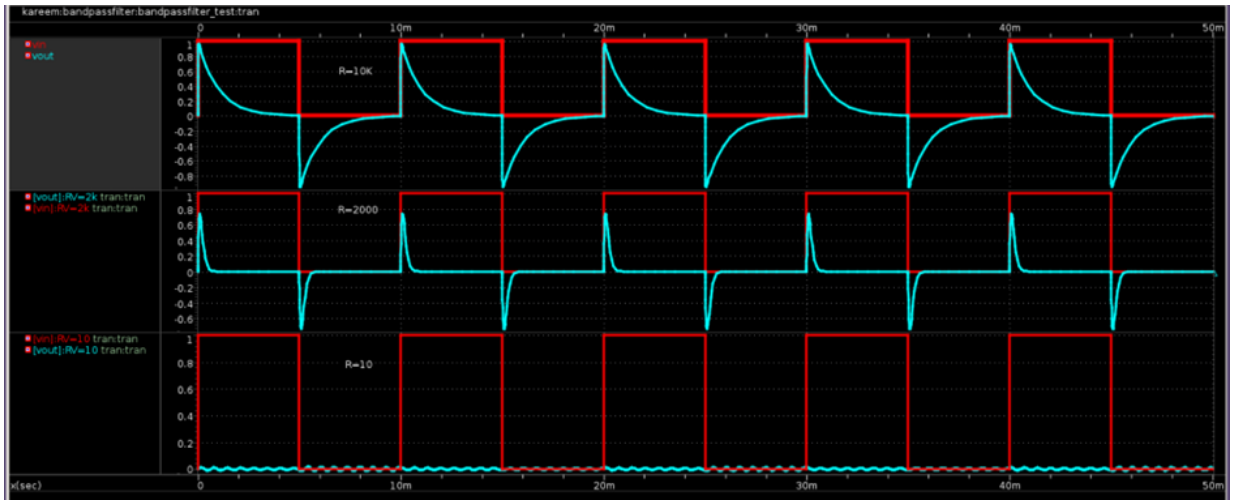


Figure 14: Parametric sweep overlay — underdamped ($R = 10 \Omega$), critically damped ($R = 2 \text{ k}\Omega$), overdamped ($R = 10 \text{ k}\Omega$).